# Exponential Tangential Advantages of Noncommuting Matrix-Polynomial and Linear-Phase MIMO FIR Filters

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**Abstract**

Scalar Chebyshev filtering treats every vector in a block Krylov iterate with the same polynomial, while simultaneously diagonalizable matrix coefficients amount to independent scalar filters after a fixed channel rotation. We study the larger class of Hermitian matrix-polynomial filters under tangential pass constraints. Our main result gives an asymptotic separation: at degree $N$ and channel dimension $d = N^2$, there are distinct pass points and full-spark channel signatures for which a symmetric noncommuting filter has stopband leakage $O(e^{-cN})$, whereas every feasible pairwise-commuting symmetric filter is identically the identity and has leakage one. An affine cosine substitution turns the theorem into a fixed-latency separation for reciprocal linear-phase MIMO FIR filters: fully coupled taps can achieve exponential high-frequency rejection, while every feasible fixed orthogonal bank of scalar filters is a pure delay. We also give an exact rational five-tap certificate with stopband norm at most 301/304 and prove that its advantage survives commuting calibration errors $\delta < 3/32300$. A second exact-arithmetic certificate is calibrated from a public triaxial pump-vibration data set: it gives leakage below 0.96 versus one for the exactly calibrated commuting class, while its maximum directional error on six held-out records is 0.00385. In a frequency-domain-decomposition test on the frozen held-out cospectra, filtering rotates the leading modal direction by at most 0.222° and changes its leading spectral ordinate by at most $2.4 \times 10^{-5}$ relatively. Numerical programs test the mechanism and expose an ordinary graph-denoising setting in which scalar Chebyshev filtering is instead preferable. Tangential matrix interpolation, MIMO filtering and convex FIR design are not claimed as new; priority of the precise fixed-order separation remains under audit.

## 1 Problem and contribution

Let

$$P(x) = \sum_{k=0}^{N} C_k x^k, \qquad C_k \in \mathbb{R}^{d\times d},$$

and let $I_- = [-1, 0]$ be a stopband. Given distinct pass points $\lambda_j > 0$ and nonzero channel signatures $v_j \in \mathbb{R}^d$, impose

$$P(\lambda_j)v_j = v_j, \qquad 1 \le j \le M. \tag{1}$$

The leakage is

$$\rho(P) = \sup_{x\in I_-} \|P(x)\|_{\mathrm{op}}. \tag{2}$$

The coefficients will always be real symmetric. They need not commute.

The comparator is the pairwise-commuting subclass. Such coefficients admit a common orthogonal eigenbasis, so a fixed channel rotation turns $P$ into a diagonal collection of scalar filters. The question is whether allowing the channel mixing to depend on the polynomial degree can reduce (2) under the same pass constraints.

Our answer is exponentially affirmative in an existence sense. The input signatures are full spark: every $d$ of them span $\mathbb{R}^d$. This property makes the commuting obstruction exact, while an arbitrarily small, explicitly chosen perturbation of a coordinate construction retains a noncommuting interpolant with exponentially small leakage.

# 2 Main separation theorem

**Theorem 2.1** (Exponential tangential separation)**.** *There are absolute constants $C, c > 0$ and $N_0 \in \mathbb{N}$ such that for every $N \geq N_0$, with $d = N^2$ and $M = d + N$, one can choose distinct rational pass points $\lambda_1, \ldots, \lambda_M \in [1, 7/2]$ and full-spark unit vectors $v_1, \ldots, v_M \in \mathbb{R}^d$ with the following properties.*

1. *There is a degree-$N$ polynomial $P_N$ with real symmetric coefficient matrices satisfying* (1) *and*
$$\rho(P_N) \leq Ce^{-cN}.$$
2. *If a degree-$N$ polynomial $Q_N$ has pairwise-commuting real symmetric coefficient matrices and satisfies the same tangential constraints, then $Q_N(x) = I_d$ for every $x$. In particular, $\rho(Q_N) = 1$.*

*For all sufficiently large $N$, the coefficient matrices of $P_N$ cannot commute pairwise.*

We split the proof into an obstruction, a small-leakage base construction, and a stable full-spark perturbation.

## 2.1 The commuting obstruction

**Lemma 2.2.** *Let $M \geq d + N$, let the $\lambda_j$ be distinct, and let the $v_j \in \mathbb{R}^d$ be full spark. If $Q(x) = \sum_{k=0}^{N} D_k x^k$ has pairwise-commuting real symmetric coefficients and $Q(\lambda_j)v_j = v_j$ for every $j$, then $Q = I_d$.*

*Proof.* The matrices $D_k$ have a common orthonormal eigenbasis $u_1, \ldots, u_d$. Write $q_r(x)$ for the scalar degree-$N$ polynomial induced by $Q(x)$ on $u_r$. Taking the $u_r$ component of the $j$th interpolation constraint gives
$$(q_r(\lambda_j) - 1)\langle u_r, v_j\rangle = 0.$$
A nonzero vector can be orthogonal to at most $d - 1$ of the full-spark targets: otherwise $d$ targets would lie in the $(d-1)$-dimensional hyperplane $u_r^{\perp}$ and could not span $\mathbb{R}^d$. Thus at least $M - d + 1 \geq N + 1$ distinct values of $\lambda_j$ are roots of $q_r - 1$. Consequently $q_r = 1$. This holds for every $r$, so $Q = I_d$. □

## 2.2 A diagonal base filter

Let $d = N^2$. Partition $M = d + N$ nodes into $d$ coordinate groups: the first $N$ groups contain two nodes each and the remaining groups contain one. One concrete choice, with $1 \leq r \leq N$ and $N < r \leq d$, is

$$a_r = 1 + \frac{r}{10N}, \qquad b_r = 2 + \frac{r}{10N}, \tag{3}$$
$$c_r = 3 + \frac{r - N}{2d}. \tag{4}$$

These nodes are distinct, lie in $[1, 7/2]$, and have mutual separation at least $(2N^2)^{-1}$.

Write $T_m$ for the Chebyshev polynomial of the first kind and set $\phi(x) = 2x + 1$, which maps $I_-$ onto $[-1, 1]$. For a one-node group with node $\lambda$, define

$$q_\lambda(x) = \frac{T_N(\phi(x))}{T_N(\phi(\lambda))}. \tag{5}$$

For a two-node group with nodes $a \neq b$, define

$$q_{a,b}(x) = \frac{x-b}{a-b}\frac{T_{N-1}(\phi(x))}{T_{N-1}(\phi(a))} + \frac{x-a}{b-a}\frac{T_{N-1}(\phi(x))}{T_{N-1}(\phi(b))}. \tag{6}$$

These polynomials equal one at their assigned node or nodes. Put them on the diagonal of $P_0(x)$ according to the coordinate partition.

**Lemma 2.3.** *Let*

$$\eta_0 = \operatorname{arcosh}(3), \qquad \eta_1 = \operatorname{arcosh}(8), \qquad \Delta\eta = \eta_1 - \eta_0.$$

*Then*

$$\sup_{x \in I_-} \|P_0(x)\|_{\mathrm{op}} \leq \frac{52}{5} e^{\eta_0} e^{-\eta_0 N}.$$

*Moreover, at every pass node $\lambda_j \in [1, 7/2]$, $\|P_0(\lambda_j)\|_{\mathrm{op}} \leq 10 e^{N\Delta\eta}$ uniformly in the group labels.*

*Proof.* For $x \in I_-$, $|T_m(\phi(x))| \leq 1$. For $\lambda \geq 1$,

$$T_m(\phi(\lambda)) = \cosh(m \operatorname{arcosh}(2\lambda + 1)) \geq \tfrac{1}{2} e^{m\eta_0}, \quad \eta_0 = \operatorname{arcosh}(3).$$

The one-point stopband bound is $2e^{-N\eta_0}$. In (3), $b_r - a_r = 1$. On $I_-$, the sum of the absolute Lagrange factors in (6) is at most $31/10 + 21/10 = 26/5$. The two-point bound is therefore $(52/5)e^{-(N-1)\eta_0}$.

At a pass node, $2\lambda + 1 \leq 8$, so a Chebyshev ratio is at most $2e^{N\Delta\eta}$. For the paired construction, the sum of the absolute Lagrange factors is at most five. This gives the stated pass-node bound. The operator norm of a diagonal polynomial is its largest entry magnitude. $\square$

### 2.3 An explicit full-spark perturbation

Index all pass nodes by $j = 1, \ldots, M$, let $g(j)$ be the coordinate group of node $j$, and set $t_j = j/(M+1)$. Define

$$w_j = (1, t_j, \ldots, t_j^{d-1})^\top, \quad \varepsilon_N = e^{-N^3}, \quad \widetilde{v}_j = e_{g(j)} + \varepsilon_N w_j, \quad v_j = \frac{\widetilde{v}_j}{\|\widetilde{v}_j\|_2}. \tag{7}$$

**Lemma 2.4.** *The vectors in* (7) *are full spark.*

*Proof.* Fix any $d$ indices. The determinant with the corresponding unnormalized vectors as columns is a polynomial in $\varepsilon_N$ with rational coefficients. Its coefficient at degree $d$ is the Vandermonde determinant of the corresponding $w_j$, which is nonzero because their $t_j$ are distinct. Thus the determinant polynomial is not zero. By the Lindemann–Weierstrass theorem, $e^{-N^3}$ is transcendental, so it cannot be a root of a nonzero polynomial with rational coefficients. Column normalization does not change whether the determinant vanishes. $\square$

**Lemma 2.5** (Stable symmetric interpolation)**.** *For all sufficiently large $N$, there is a real-symmetric coefficient polynomial $P_N$ satisfying $P_N(\lambda_j)\widetilde{v}_j = \widetilde{v}_j$ and*

$$\sup_{x \in I_-} \|P_N(x) - P_0(x)\|_{\mathrm{op}} \leq 64152\, N^9 \exp(-N^3 + N\Delta\eta).$$

*Proof.* Let $\mathcal{S}_N$ be the finite-dimensional space of degree-at-most-$N$ polynomials with real symmetric coefficients, and define

$$L_\varepsilon : \mathcal{S}_N \longrightarrow (\mathbb{R}^d)^M, \qquad L_\varepsilon(P) = \big(P(\lambda_j)(e_{g(j)} + \varepsilon w_j)\big)_{j=1}^M.$$

At $\varepsilon = 0$, this map is onto. Indeed, for an off-diagonal entry $p_{rs} = p_{sr}$, arbitrary output data prescribe values at the union of nodes assigned to coordinates $r$ and $s$. That union contains at most four nodes. For $N \geq 3$, ordinary Lagrange interpolation is onto on those nodes. A diagonal entry sees at most two nodes.

Equip the output space with the maximum Euclidean block norm and $\mathcal{S}_N$ with

$$\|P\|_X = \max\left\{ \sup_{x \in I_-} \|P(x)\|_{\mathrm{op}}, \max_j \|P(\lambda_j)\|_{\mathrm{op}} \right\}.$$

The entrywise construction gives a right inverse $R_N$. For $N \geq 5$, the node separation is at least $(2N^2)^{-1}$, while any two points in the stop/pass domain are at distance at most $9/2$. A Lagrange basis polynomial on at most four nodes is bounded by $(9N^2)^3$. Summing at most four basis polynomials and using $\|A\|_{\mathrm{op}} \leq d \max_{r,s} |A_{rs}|$ gives

$$\|R_N\|_{Y \to X} \leq 4(9N^2)^3 d = 2916 N^8. \tag{8}$$

Write $L_\varepsilon = L_0 + \varepsilon K_N$. Since $\|w_j\|_2 \leq \sqrt{d} = N$, $\|K_N\|_{X \to Y} \leq N$. Hence

$$L_\varepsilon R_N = I + E_N, \qquad \|E_N\| \leq 2916 \varepsilon N^9.$$

For $\varepsilon = \varepsilon_N$ and large $N$, the inverse $(I + E_N)^{-1}$ exists and has norm at most two.

Since $L_0(P_0) = (e_{g(j)})_j$, the residual at the perturbed targets is

$$r_N = (\widetilde{v}_j)_j - L_{\varepsilon_N}(P_0) = \varepsilon_N \big((I - P_0(\lambda_j)) w_j\big)_j.$$

By Lemma 2.3,

$$\|r_N\|_Y \leq \varepsilon_N N (1 + 10 e^{N\Delta\eta}) \leq 11 \varepsilon_N N e^{N\Delta\eta}.$$

Set

$$\Delta P_N = R_N (I + E_N)^{-1} r_N, \qquad P_N = P_0 + \Delta P_N.$$

Then $L_{\varepsilon_N}(P_N) = (\widetilde{v}_j)_j$. Using (8) and $\|(I + E_N)^{-1}\| \leq 2$ gives

$$\|\Delta P_N\|_X \leq 2(2916 N^8)(11 \varepsilon_N N e^{N\Delta\eta}) = 64152 N^9 e^{-N^3 + N\Delta\eta}.$$

Every coefficient remains real symmetric because $R_N$ maps into $\mathcal{S}_N$. □

*Proof of Theorem 2.1.* The normalized and unnormalized interpolation constraints are equivalent. Combine Lemmas 2.3, 2.4, and 2.5. Choose $N_0 \geq 5$ so that $2916 e^{-N^3} N^9 \leq 1/2$ and the correction in Lemma 2.5 is at most $(52/5) e^{\eta_0} e^{-\eta_0 N}$ for $N \geq N_0$. The first conclusion holds with

$$C = \frac{104}{5} e^{\eta_0}, \qquad c = \eta_0.$$

The second conclusion is Lemma 2.2. For large $N$, the first leakage is below one, so the constructed coefficients cannot commute pairwise. □

## 3 Block-Krylov meaning

The theorem has a direct interpretation for block polynomial filtering. Let $H = \sum_j \lambda_j q_j q_j^\top$ be real symmetric and let $B \in \mathbb{R}^{n\times d}$ be a starting block. For $P(x) = \sum_{k=0}^N C_k x^k$, define the right-coefficient block filter

$$\mathfrak{F}_P(H,B) = \sum_{k=0}^{N} H^k B C_k. \tag{9}$$

Its $j$th spectral row is

$$q_j^\top \mathfrak{F}_P(H,B) = (q_j^\top B)P(\lambda_j). \tag{10}$$

**Proposition 3.1** (Pass preservation and stop contraction)**.** *Let $\mathcal{T}$ and $\mathcal{S}$ index target and stop eigenpairs. Assume the rows $b_j^\top = q_j^\top B$ obey*

$$b_j^\top P(\lambda_j) = b_j^\top \quad (j \in \mathcal{T}), \qquad \lambda_j \in I_- \quad (j \in \mathcal{S}).$$

*Then the target spectral component is preserved exactly, and*

$$\|Q_\mathcal{S}^\top \mathfrak{F}_P(H,B)\|_F \le \rho(P)\|Q_\mathcal{S}^\top B\|_F,$$

*where $Q_\mathcal{S}$ has columns $q_j$, $j \in \mathcal{S}$. If the $C_k$ are pairwise-commuting symmetric matrices, one fixed orthogonal channel rotation reduces* (9) *to $d$ independent scalar polynomial filters.*

*Proof.* The pass statement follows from (10). For the stop part,

$$\|Q_\mathcal{S}^\top \mathfrak{F}_P(H,B)\|_F^2 = \sum_{j\in\mathcal{S}} \|b_j^\top P(\lambda_j)\|_2^2 \le \rho(P)^2 \sum_{j\in\mathcal{S}} \|b_j\|_2^2.$$

For the final claim, simultaneously diagonalize all $C_k = U\Lambda_k U^\top$. Then the columns of $BU$ are acted on by the scalar polynomials whose coefficients are the diagonal entries of the $\Lambda_k$. □

Because every coefficient in Theorem 2.1 is symmetric, its column constraints transpose to the row constraints in Proposition 3.1. The theorem therefore gives block spectral instances with exponentially small stop contraction that no fixed-basis collection of scalar pass-preserving filters can reproduce. This is an approximation-theoretic separation; an end-to-end runtime advantage also depends on filter construction, recurrence stability, orthogonalization, and the cost model.

## 4 Reciprocal linear-phase MIMO FIR meaning

The polynomial separation is equivalently a fixed-order statement for a standard multichannel FIR class. Let

$$x(y) = \frac{9y+5}{4}, \qquad G_N(\omega) = P_N(x(\cos\omega)).$$

Because $G_N$ is a real symmetric matrix cosine polynomial of degree $N$, it has a unique expansion

$$G_N(\omega) = A_N + 2\sum_{r=1}^{N} A_{N-r}\cos(r\omega), \qquad A_r = A_r^\top.$$

The causal response

$$H_N(e^{i\omega}) = \sum_{r=0}^{2N} A_r e^{-ir\omega} = e^{-iN\omega}G_N(\omega), \qquad A_r = A_{2N-r}, \tag{11}$$

therefore has $2N+1$ real symmetric palindromic taps and common group delay $N$.

**Corollary 4.1** (Fixed-latency MIMO FIR separation). *For every sufficiently large $N$ there are $M = N^2 + N$ distinct frequencies $\omega_j \in [0, \arccos(-1/9)]$ and full-spark directions $v_j \in \mathbb{R}^{N^2}$ for which a causal response of the form* (11) *satisfies*

$$H_N(e^{i\omega_j})v_j = e^{-iN\omega_j}v_j, \qquad \sup_{\omega\in[\arccos(-5/9),\pi]} \|H_N(e^{i\omega})\|_{\mathrm{op}} \le Ce^{-cN}.$$

*Every response with the same real-symmetric palindromic tap structure whose taps commute pairwise and which satisfies the same calibrations is the pure delay $e^{-iN\omega}I$. Its high-frequency norm is one.*

*Proof.* Take the data of Theorem 2.1 and choose $\cos\omega_j = (4\lambda_j - 5)/9$. The pass interval $[1, 7/2]$ maps into $[-1/9, 1]$, while the stop interval $[-1, 0]$ maps onto $[-1, -5/9]$. Equation (11) gives the calibration and transfers the norm bound unchanged. The affine substitution is invertible on coefficient spaces, as is the change between powers of $\cos\omega$ and the cosine basis. Thus the transformed coefficient matrices commute pairwise exactly when the original polynomial coefficients do. Palindromy makes the FIR taps an invertible relabelling and rescaling of the cosine coefficients. Applying Lemma 2.2 therefore forces the zero-phase response to be $I$. $\square$

A fixed orthogonal channel rotation diagonalizes commuting symmetric taps, so the comparator in Corollary 4.1 is precisely a bank of independent scalar linear-phase filters in one frequency-independent channel basis. The result does not compare implementation cost, quantization robustness, or unconstrained MIMO filters, and the calibrated directions are constructed rather than taken from a deployed system.

# 5 Exact finite witnesses

## 5.1 A full-spark quadratic certificate

The asymptotic mechanism already occurs with rational data at degree two. Take

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = \left(\tfrac{1}{2}, 1, \tfrac{3}{2}, 2\right), \qquad (v_1, v_2, v_3, v_4) = \left(\begin{pmatrix}1\\-2\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}\right).$$

Every pair of targets is linearly independent, hence the tuple is full spark in $\mathbb{R}^2$. Set $P(x) = C_0 + xC_1 + x^2C_2$, with

$$C_0 = \begin{pmatrix} 2/5 & -3/16 \\ -3/16 & 29/32 \end{pmatrix}, \quad C_1 = \begin{pmatrix} 15/16 & 3/20 \\ 3/20 & 3/32 \end{pmatrix}, \quad C_2 = \begin{pmatrix} -3/8 & 0 \\ 0 & -3/80 \end{pmatrix}.$$

Direct substitution verifies $P(\lambda_j)v_j = v_j$ for all four targets, and

$$[C_0, C_1] = \begin{pmatrix} 0 & 1053/12800 \\ -1053/12800 & 0 \end{pmatrix} \ne 0.$$

**Proposition 5.1** (Exact quadratic gap). *The polynomial above obeys*

$$\sup_{x\in[-1,0]} \|P(x)\|_{\mathrm{op}} \le \frac{301}{304} < 1.$$

*Every pairwise-commuting real-symmetric quadratic polynomial satisfying the same four tangential constraints is identically $I_2$ and has leakage one.*

*Proof.* For the upper bound, put $x = t - 1$ and express the leading principal minor and determinant of $I_2 \pm P(t-1)$ in the Bernstein basis on $t \in [0,1]$. The leading-minor coefficients are

$$\left(\tfrac{153}{80}, \tfrac{171}{160}, \tfrac{3}{5}\right), \qquad \left(\tfrac{7}{80}, \tfrac{149}{160}, \tfrac{7}{5}\right),$$

and the determinant coefficients are, respectively,

$$\begin{gathered}\left(\tfrac{81}{256}, \tfrac{1701}{10240}, \tfrac{219}{2560}, \tfrac{441}{10240}, \tfrac{27}{1280}\right), \\ \left(\tfrac{53}{1280}, \tfrac{8389}{10240}, \tfrac{783}{512}, \tfrac{21913}{10240}, \tfrac{3371}{1280}\right).\end{gathered}$$

They are all positive, so both matrices are positive definite throughout the stopband. Their trace Bernstein coefficients give the upper bounds 171/80 and 529/160. For a positive 2 by 2 matrix $S$, $\lambda_{\min}(S) \geq \det(S)/\operatorname{tr}(S)$. Consequently

$$I_2 - P(x) \succeq \frac{3}{304} I_2, \qquad I_2 + P(x) \succeq \frac{53}{4232} I_2 \succeq \frac{3}{304} I_2,$$

which proves the norm bound.

For the commuting class, apply Lemma 2.2 with $M = 4 = d + N$. Every scalar common-eigenbasis entry has three distinct roots of $q - 1$ and is identically one. □

The gap is stable under imperfect calibration. Write $\widehat{v}_j = v_j / \|v_j\|_2$.

**Proposition 5.2** (Robust commuting obstruction)**.** *Let $Q$ be a pairwise-commuting real-symmetric quadratic polynomial satisfying*

$$\|Q(\lambda_j)\widehat{v}_j - \widehat{v}_j\|_2 \leq \delta, \qquad j = 1, \ldots, 4.$$

*Then*

$$\sup_{x \in [-1,0]} \|Q(x)\|_{\mathrm{op}} \geq 1 - \frac{425}{4}\delta.$$

*In particular, if $\delta < 3/32300$, its leakage is strictly larger than the certified 301/304 leakage of the noncommuting witness.*

*Proof.* For any two distinct normalized targets $a, b$ in the tuple, $|a^\top b|^2 \leq 9/10 < (593/625)^2$. For every unit vector $u$,

$$|u^\top a|^2 + |u^\top b|^2 = u^\top (aa^\top + bb^\top) u \geq 1 - |a^\top b| > \frac{32}{625}.$$

Thus at most one target has $|u^\top \widehat{v}_j| < 4/25$. Simultaneously diagonalize $Q$ and fix a common unit eigenvector $u$, with scalar quadratic entry $q$. Projecting the residual bound shows that at three distinct pass nodes

$$|q(\lambda_j) - 1| \leq \frac{25}{4}\delta.$$

For each of the four possible three-node subsets of $\{1/2, 1, 3/2, 2\}$, the sum of the absolute Lagrange weights at $x = 0$ is, respectively, $7, 5, 5$, or $17$. Hence $|q(0) - 1| \leq (425/4)\delta$. This holds for both common eigenvectors, so $\|Q(0)\|_{\mathrm{op}} \geq 1 - (425/4)\delta$. Finally, $1 - 301/304 = 3/304$ gives the stated threshold. □

The exact-arithmetic replay `verification/verify_quadratic_filter_witness.py` checks all constraints, target minors, commutators, Bernstein coefficients, and the rational robust-calibration constants without a numerical solver.

### 5.2 An exact five-tap MIMO FIR certificate

The quadratic witness has an exact causal FIR realization. Substitute $x = (3\cos\omega + 1)/2$ and define the palindromic taps

$$B_0 = B_4 = \begin{pmatrix} -27/128 & 0 \\ 0 & -27/1280 \end{pmatrix}, \quad B_1 = B_3 = \begin{pmatrix} 27/64 & 9/80 \\ 9/80 & 27/640 \end{pmatrix},$$

$$B_2 = \begin{pmatrix} 113/320 & -9/80 \\ -9/80 & 577/640 \end{pmatrix}.$$

Their response obeys the exact identity

$$\sum_{r=0}^{4} B_r e^{-ir\omega} = e^{-2i\omega}\big(B_2 + 2B_1\cos\omega + 2B_0\cos 2\omega\big) = e^{-2i\omega} P\left(\frac{3\cos\omega + 1}{2}\right).$$

Consequently it preserves the four directions, up to their common delay phase, at cosine values $0, 1/3, 2/3, 1$ and has operator norm at most $301/304$ on $\omega \in [\arccos(-1/3), \pi]$. Every commuting symmetric palindromic five-tap response meeting the same four calibrations is the pure two-sample delay and has norm one. Moreover $[B_0, B_1] \neq 0$. All statements, including the tap conversion, are checked by the same exact-arithmetic replay. The same change of variables transfers Proposition 5.2: if a commuting five-tap comparator obeys $\|e^{2i\omega_j} H(e^{i\omega_j})\widehat{v}_j - \widehat{v}_j\|_2 \le \delta$, its high-frequency norm is at least $1 - (425/4)\delta$.

### 5.3 An affine two-constraint precursor

For degree one and $d = 2$, take pass data

$$\lambda_1 = \tfrac{1}{2}, \quad v_1 = (1,0)^\top, \qquad \lambda_2 = 1, \quad v_2 = (1,1)^\top,$$

and $P(x) = A + xB$ with

$$A = \begin{pmatrix} 4/5 & 1/5 \\ 1/5 & 3/10 \end{pmatrix}, \qquad B = \begin{pmatrix} 2/5 & -2/5 \\ -2/5 & 9/10 \end{pmatrix}.$$

The constraints hold exactly and

$$[A, B] = \begin{pmatrix} 0 & -1/10 \\ 1/10 & 0 \end{pmatrix}.$$

Convexity of $x \mapsto \|A + xB\|_{\mathrm{op}}$ puts its maximum over $I_-$ at an endpoint. Direct diagonalization gives

$$\rho(P) = \max\{\|A\|_{\mathrm{op}}, \|A - B\|_{\mathrm{op}}\} = \frac{1 + \sqrt{61}}{10} < 0.882.$$

If $A$ and $B$ commute, their common eigenbasis contains a direction with nonzero overlap with both nonorthogonal, nonparallel targets. Its scalar affine entry equals one at two points and hence everywhere. The commuting leakage is at least one.

## 6 Numerical stress test

The repository script `research/route_c_scaling_pilot.py` solves a grid SDP with symmetric coefficient variables. For degree 3, dimension 9, 12 perturbed targets and seed 7, the results are:

| Perturbation | SDP objective | Validation maximum | Commutator norm |
|---|---|---|---|
| 0.001 | 0.2983490 | 0.2983682 | 0.466925 |
| 0.010 | 0.3199793 | 0.3200281 | 0.391179 |
| 0.030 | 0.3977121 | 0.3977588 | 0.708604 |

The optimization uses 81 stopband points and validation uses 4001 independent points. Clarabel labels these solves `optimal_inaccurate`; they are diagnostics, not proof. The exact witnesses and Theorem 2.1 do not depend on solver output.

### 6.1 Controlled block-iteration benchmark

The script `research/route_c_block_krylov_benchmark.py` embeds the quadratic certificate in a 256-dimensional Hermitian eigenproblem. The four pass eigenvalues are $1/2, 1, 3/2, 2$, the other 252 eigenvalues fill $[-1, 0]$, and the initial stop/pass Frobenius ratio is five. We repeatedly apply the degree-two matrix filter and compare it with

$$q(x) = \frac{8x^2 + 8x + 1}{7},$$

the scalar degree-two Chebyshev stopband filter normalized only at $x = 1/2$. Let $\sin\theta$ denote the largest principal-angle sine from the filtered block to the clean two-dimensional block defined by the four prescribed pass rows. The scalar pass error below is minimized over one global rescaling, so it does not merely report scalar amplitude growth.

| Iterations | Matrix stop/pass | Matrix $\sin\theta$ | Scalar pass error | Scalar $\sin\theta$ |
|---|---|---|---|---|
| 0 | 5.0000 | 0.9892 | 0.0000 | 0.9892 |
| 1 | 3.5776 | 0.9719 | 0.5616 | 0.5568 |
| 5 | 2.4352 | 0.9458 | 0.7832 | 0.9727 |
| 20 | 1.0318 | 0.7755 | 0.8018 | 0.9871 |
| 100 | 0.0443 | 0.0525 | 0.8018 | 0.9982 |

The matrix pass error stays below $6 \times 10^{-16}$. The scalar filter has stopband maximum $1/7$ and initially suppresses contamination much faster, but its pass multipliers are $(1, 17/7, 31/7, 7)$; iteration consequently rotates the clean block away from the prescribed one. The fair commuting comparator under all four exact tangential constraints is the identity and remains at stop/pass ratio 5 and $\sin\theta = 0.9892$. This controlled construction demonstrates the theorem's error mechanism, not a runtime advantage on a natural problem.

### 6.2 A measured triaxial calibration and exact continuum certificate

We next use VBL-VA001, an open lab-scale pump-vibration data set with triaxial acceleration records sampled at 20 kHz [2, 11]. The task is frequency-domain-decomposition (FDD) preprocessing, not fault classification. FDD extracts mode-shape estimates as leading singular vectors of cross-spectral matrices at spectral peaks [6]; filter-induced perturbations are therefore a scientifically relevant failure mode, and filtering artifacts can corrupt modal estimates [20]. We selected twelve normal-condition records by the archive prefixes `normal_000` and `normal_001`, using all six records under the first prefix for fitting and retaining all six under the second prefix. The file names, CRC-32 and SHA-256 digests were frozen before optimization.

Each record was polyphase-decimated by eight, then divided into length-1024 Hann windows with 50% overlap. We averaged the real 3 by 3 cospectral matrix and, using the training split only, selected the five largest-trace peaks between 20 and 1000 Hz subject to an eight-bin separation. Their leading cospectral eigenvectors are the channel signatures. The frequency node is $x = \cos(2\pi f/2500)$. Table 1 reports the six-decimal rationalization used by the exact certificate; every displayed component and node differs from its measured value by less than $5 \times 10^{-7}$.

**Lemma 6.1** (Modal-component invariance)**.** *Let the cospectral matrix at a modal frequency have the decomposition $G_j = \lambda_j v_j v_j^\top + R_j$, where $\lambda_j \geq 0$, $\|v_j\|_2 = 1$, and $R_j \succeq 0$. If a real-symmetric response $P$ obeys $P(x_j)v_j = v_j$, then the reciprocal palindromic FIR response $H(e^{\mathrm{i}\omega}) = e^{-\mathrm{i}N\omega} P(\cos\omega)$ obeys*

$$H(e^{\mathrm{i}\omega_j}) G_j H(e^{\mathrm{i}\omega_j})^* = \lambda_j v_j v_j^\top + P(x_j) R_j P(x_j).$$

*Thus the rank-one modal component, its channel shape, and its amplitude are preserved exactly, up to one common $N$-sample delay, while the residual cospectrum is filtered.*

*Proof.* The scalar phase cancels in $HG_jH^*$, and symmetry gives $P(x_j)v_j v_j^\top P(x_j) = v_j v_j^\top$. □

| $f$ (Hz) | $x$ | $v_x$ | $v_y$ | $v_z$ |
|---|---|---|---|---|
| 100.0977 | 0.968522 | −0.054531 | 0.996129 | −0.068944 |
| 200.1953 | 0.876070 | 0.980644 | 0.167125 | 0.102012 |
| 744.6289 | −0.296151 | 0.152303 | 0.986844 | −0.054250 |
| 844.7266 | −0.524590 | 0.869001 | −0.350335 | 0.349432 |
| 944.8242 | −0.720003 | 0.606651 | 0.716228 | −0.344951 |

Table 1: Training frequencies and rationalized triaxial signatures from VBL-VA001.

**Proposition 6.2** (Data-grounded rational certificate)**.** *For the five rational nodes and signatures in Table 1, there is a real-symmetric quadratic polynomial $P$ satisfying $P(x_j)v_j = v_j$ exactly and*

$$[C_k, C_\ell] \neq 0 \quad \textit{for some } k, \ell, \qquad \sup_{-1 \leq x \leq -929/1000} \|P(x)\|_{\mathrm{op}} < \frac{24}{25}.$$

*Every pairwise-commuting real-symmetric quadratic satisfying the same five calibrations is identically $I_3$ and has leakage one.*

*Proof.* All ten 3 by 3 target determinants are nonzero in exact rational arithmetic; the smallest normalized absolute determinant is $0.009489456048\ldots$. Lemma 2.2, with $d = 3$, $N = 2$ and $M = 5$, therefore proves the commuting conclusion.

For the upper bound, fix the $(1,1)$, $(1,2)$ and $(2,2)$ entries of $C_2$ to $-0.316171$, $-0.537470$ and $-0.909040$, respectively, and solve the remaining 15 rational linear equations for the other symmetric entries. The exact replay verifies the calibrations and a nonzero commutator. On the 101-point rational grid of $[-1, -929/1000]$, Sylvester's criterion gives

$$(191/200)^2 I_3 - P(x_i)^2 \succ 0.$$

The same replay proves $\sup_{|x| \leq 1} \|P'(x)\|_{\mathrm{op}} \leq 5.740500963$. The grid spacing is $71/100000$, hence every point is within half a grid spacing of some $x_i$ and

$$\|P(x)\|_{\mathrm{op}} < \frac{191}{200} + 5.740500963 \frac{71}{200000} < 0.957038 < \frac{24}{25}.$$

Every comparison in this paragraph is performed on integers and rational fractions; the printed decimals only abbreviate upward-valid bounds. □

The held-out signatures lie between 0.129° and 0.460° from their training counterparts. Without refitting, the rational witness has maximum held-out residual 0.0038491 and dense-grid leakage 0.94870. The numerical comparators in Table 2 use the unrounded measured nodes and signatures. A general nonsymmetric MIMO polynomial attains much lower training leakage, but its held-out residual is 0.1045 and its largest coefficient Frobenius norm is 36.06, compared with 2.623 for the rational symmetric witness. This single split suggests a reciprocity–robustness tradeoff; it does not establish a statistical generalization theorem.

For a task-level replay, we froze the full 3 by 3 training and held-out cospectral matrices at the five peaks, applied $P(x_j)G_jP(x_j)$ on the held-out split, and recomputed the FDD leading eigenpair. Across the five frequencies, the angle between the unfiltered and filtered leading directions is at most 0.2217°; the leading eigenvalue ratio differs from one by at most $2.40 \times 10^{-5}$. These figures are outputs of the offline replay, not consequences of the exact certificate. Lemma 6.1 explains the specification: exact calibration preserves an ideal modal component, while the held-out calculation measures the effect of residual components and train–test drift.

| Filter | Stop leakage | Fit tolerance | Held-out residual |
|---|---|---|---|
| Rational symmetric witness | $< 0.96000$ | exact | 0.003849 |
| Symmetric SDP optimum | 0.94867 | exact | 0.003849 |
| General nonsymmetric SDP | 0.14514 | exact | 0.10447 |
| Scalar quadratic optimum | 0.97520 | 0.01 | 0.01000 |
| Best of 64 fixed bases | 0.96966 | 0.01 | 0.01011 |
| Exact commuting class | 1 | exact | 0 |

Table 2: Measured-calibration controls. The 64-basis row is a seeded stress test, not a global optimization certificate over orthogonal bases.

Thus the measured instance closes both the purely constructed-data objection and the objection that tangential preservation has no task meaning: its calibration and validation use disjoint sensor records, FDD supplies the mode-shape objective, and the exact 0.96-versus-1 gap is not solver-dependent. It does not show improved fault diagnosis, damping estimation, hardware cost or deployment performance; those claims are outside the present experiment.

### 6.3 A failed graph-denoising application gate

We also tested the standard MIMO graph-filter form $Y = \sum_{k=0}^{2} S^k X C_k$ on two smooth fields over an 18 by 23 grid graph, using four low-frequency full-spark row signatures. The symmetric MIMO SDP collapsed to the identity (validated stop norm 0.99999992), while the one-point-normalized scalar Chebyshev filter had stop norm 1/17. After two iterations its best globally rescaled pass error was only 0.00524 and its desired-subspace sine was 0.0292, compared with 0.997 for the symmetric MIMO solution. A nonsymmetric MIMO fit eventually preserved the exact geometry, but was inferior at short iteration counts. Thus ordinary graph denoising does not justify the exact tangential specification, and no such application claim is made. We retain this negative result to prevent selecting an example merely because it matches the theorem's notation.

## 7 Relation to established work and open gates

Tangential matrix-polynomial interpolation is classical [4, 7]. Structured matrix-polynomial interpolation with Hermitian, positive-semidefinite and related symmetries is also established [1]; that work treats full matrix values and lists the tangential structured variant as a future problem. Matrix-valued Nevanlinna–Pick theory handles supremum-norm analytic interpolation, including complexity constraints [5, 3]. Kuroiwa and Lindquist treat the bitangential problem with explicit McMillan-degree constraints [14]. Most directly, Stefanovski and Georgijević prove that real stable rational matrices can meet bitangential constraints while having arbitrarily small spectral norm on a proper frequency region [17]; their degree grows and they do not compare commuting and noncommuting Hermitian polynomial coefficients. Thus small regional norm under matrix tangential constraints is not itself our contribution. Tangential interpolation is also a standard bridge between MIMO model reduction and Krylov projection [8]; modern block rational approximation includes matrix-valued weights, noisy-data comparisons, and tangential interpolation mechanisms [10]. Recent iterative MIMO schemes even select low-rank tangential data using maximum-error criteria [12]. Their maximum-error step selects the next interpolation frequency, whereas their proved monotonicity concerns a weighted $H_2$ error; monotone $H_\infty$ bounds are stated as future work. Gallivan et al.'s main result instead gives a generically unique minimal-McMillan-degree rational interpolant. Neither result provides a fixed-degree Hermitian-polynomial minimax separation from the simultaneously diagonalizable subclass. Minimax polynomial filtering and SDP formulations have a long numerical-linear-algebra history [18], and matrix-polynomial spaces arise naturally in block Krylov methods [16]. General MIMO graph filters already use matrix taps in precisely the form $\sum_k S^k X C_k$ [9], while scalar FIR minimax design by convex optimization is classical [19]. Kootsookos already formulates fixed-length MIMO FIR approximation of matrix transfer functions in the uniform $H_\infty$ norm, proves global error lower bounds, and gives MIMO and linear-phase design algorithms [13]. More closely, Ljungars and Fu optimize the operator-norm error of matrix-valued multi-channel linear-phase FIR filters on a frequency grid by SDP [15]. Their formulation has neither directional pass interpolation nor a commuting/simultaneously-diagonalizable comparator. Accordingly, fixed-length MIMO $H_\infty$ approximation, MIMO linear phase, operator-norm minimax design, and SDP are not claimed as new.

The candidate novelty is therefore narrower than norm-constrained or degree-constrained tangential interpolation or FIR design alone. Under a fixed ordinary degree—equivalently fixed reciprocal FIR latency—a Hermitian-coefficient tangential minimax problem can exhibit an exponential separation between fully coupled coefficients and the full fixed-basis scalar subclass. Before submission, these gates remain:

1. complete full-text checks of the closest polynomial, rational regional, and linear-phase MIMO FIR results; the current clause-by-clause audit is not a proof of priority;

2. obtain an external adversarial review of the proof and of the claim that the surviving conjunction is new. The FDD invariance lemma and held-out cospectral replay close the task-meaning gate at the preprocessing level, but do not constitute a deployment or diagnostic study.

**Reproducibility.** The source and research ledger are in the project repository. The repository links the exact rational replay, the block-iteration benchmark, its machine-readable output, the deliberately negative graph-filter pilot, the VBL-VA001 provenance manifest, its exact continuum replay, the offline baseline replay, and its machine-readable output, the clause-by-clause priority matrix, and the artifact manifest with the PDF hash.

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